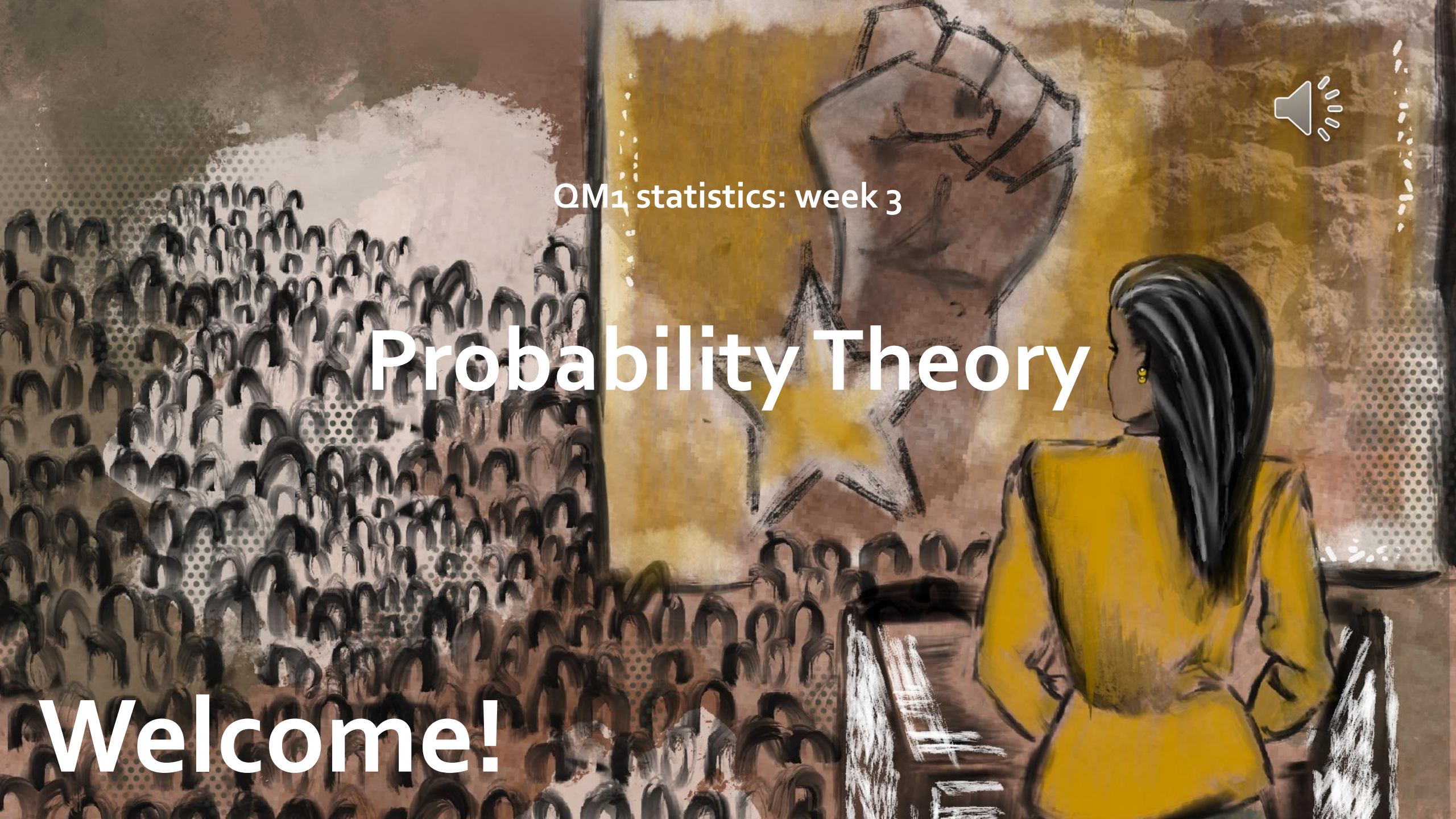
QM1 statistics: week 3

Probability Theory



Welcome!

Probability calculations

Power to the people

The Blood Arroyo drug cartel is voting for a new leader

Carlos Rubio Lagos
wants to blackmail the government into legalising cocaine

Griselda Marrón
wants to stop torturing traitors because that's inhumane (just shoot them on sight instead)

Hideki Yamaguchi
wants to bring back discipline: drug traffickers who fail their assignment should commit ritual suicide

Marrón wants to take a strategic approach, and wonders where she should campaign

		CELL				
		Anaconda	**Cascabel**	**Mamba**	**Pitón**	
TURNOUT	**voted**	385	1539	280	1746	**3950**
	did not vote	115	361	520	54	**1050**
		500	**1900**	**800**	**1800**	**5000**

- At her request, let's check how the members of the cartel behaved in the *previous* election
  - Voter turnout was not the same in every department ('cell')
- The **unconditional probability** that a randomly chosen member would have voted was

$$P(voted) = \frac{3950}{5000} = 0.79$$

- Let's label a person voting as the event *A*. Then in general, we can define the probability of any event *A* as

$$\boldsymbol{P(A)} = \frac{\#A\ (number\ of\ outcomes\ that\ fit\ this\ event)}{\#S\ (total\ number\ of\ possible\ outcomes\ a.k.a.\ \boldsymbol{sample\ space})}$$

Where should Marrón campaign? (continued)

		CELL				
		Anaconda	**Cascabel**	**Mamba**	**Pitón**	
TURNOUT	**voted**	385	1539	280	1746	**3950**
	did not vote	115	361	520	54	**1050**
		500	**1900**	**800**	**1800**	**5000**

- Her home district is Anaconda; should she stay there, or take her campaign on a tour?
- Well, at the last election, the probability that a random drug dealer would be stationed in Anaconda was pretty low:

$$P(Anaconda) = \frac{500}{5000} = 0.10$$

- Therefore, there was a high probability that dealers would be active in *a different district*
- $P(not\ Anaconda)$ is simply the <u>leftover</u>:

$$P(not\ Anaconda) = 1 - \frac{500}{5000} = \frac{4500}{5000} = 0.90$$

- We call this the **complement rule**:

$$\boldsymbol{P(A^C) = 1 - P(A)}$$

Where should Marrón campaign? (continued)

		CELL				
		Anaconda	**Cascabel**	**Mamba**	**Pitón**	
TURNOUT	**voted**	385	1539	280	1746	**3950**
	did not vote	115	361	520	54	**1050**
		500	**1900**	**800**	**1800**	**5000**

- How often did drug traffickers vote if they were from the Mamba cell?

$$P(voted|Mamba) = \frac{280}{800} = 0.35$$

Campaigning here looks like a terrible idea

- And in Pitón?

$$P(voted|Pitón) = \frac{1746}{1800} = 0.97$$

A large district with sky-high turnout: amazing!

- In general, this is the **conditional probability** of an event *A*, given another event *B*

$$\boldsymbol{P(A|B) = \frac{\#A\ in\ B}{\#B}}$$

Upon arrival in Pitón, Marrón campaigns rather aggressively

		Marrón flyer		
		accepted	turned down	
Marrón button	accepted	1680	70	1750
	turned down	210	140	350
		1890	210	2100

- Her team approaches every trafficker individually and offers them both a flyer that they can stick on their office window, and a button they can wear on their shirts. Or else.
- How likely is it that a random cartel member will have accepted both a flyer and a button?

$$P(accepted\ flyer\ and\ accepted\ button) = \frac{1680}{2100} = 0.80$$

- There are <u>no conditions</u>! This is the <u>unconditional</u> probability that two events, *A* and *B*, both happen
- Simplest approach (works with a contingency table):

$$\boldsymbol{P(A\ and\ B) = \frac{\#(A\ and\ B)}{\#S\ (total\ sample\ space)}}$$

Marrón's feisty campaign (continued)

		Marrón flyer		
		accepted	turned down	
Marrón button	accepted	1680	70	1750
	turned down	210	140	350
		1890	210	2100

- You may also use the **General Multiplication Rule** (suitable in many more situations):

$$\boldsymbol{P(A\ and\ B) = P(A) * P(B|A)}$$

- It works both ways:

$$P(accepted\ flyer) * P(acc.button|acc.flyer) = \frac{1890}{2100} * \frac{1680}{1890} = \frac{1680}{2100} = 0.800$$

$$P(accepted\ button) * P(acc.flyer|acc.button) = \frac{1750}{2100} * \frac{1680}{1750} = \frac{1680}{2100} = 0.800$$

Marrón's campaign (continued)

		Marrón flyer		
		accepted	turned down	
Marrón button	accepted	1680 — Counted twice! So subtract once	70	1750
	turned down	210	140	350
		1890	210	2100

- What's the probability that a random Pitón will accept a button <u>or</u> a flyer (or both!)?

$$\frac{1750}{2100} + \frac{1890}{2100} = \frac{3640}{2100} > 1 \text{ ???}$$

- Let's improve that calculation:

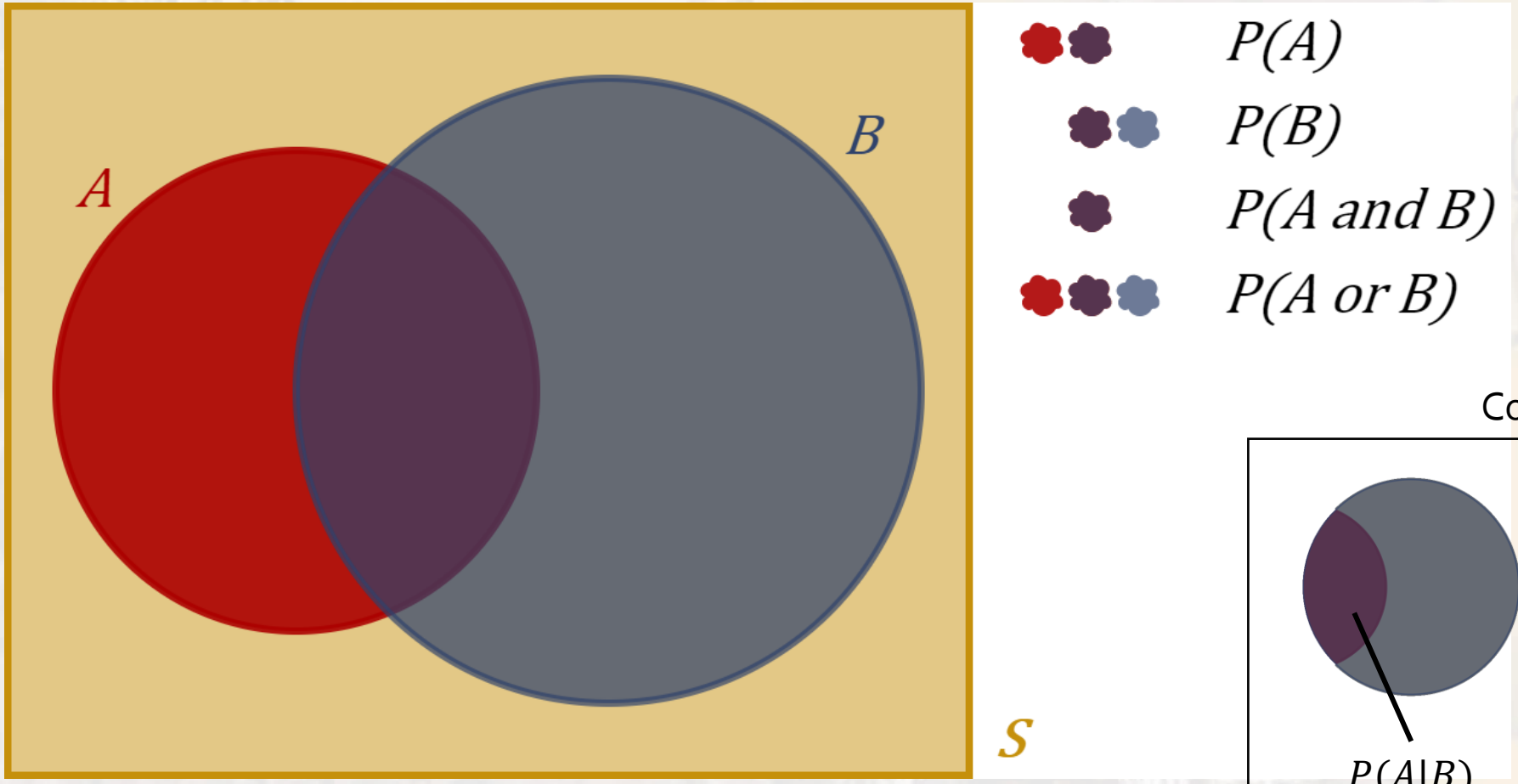
$$P(acc.flyer) + P(acc.button) - P(acc.flyer\ and\ button) = \frac{1890}{2100} + \frac{1750}{2100} - \frac{1680}{2100} = \frac{1960}{2100}$$

$$= 0.933$$

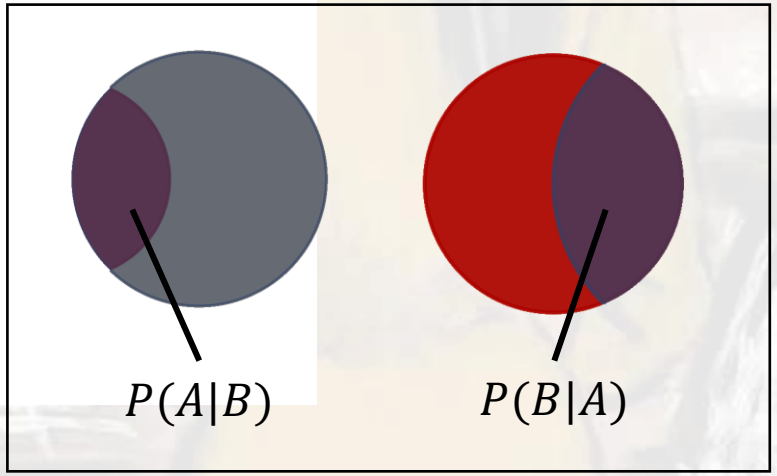
- This is then the unconditional probability that an event *A* happens, or an event *B*, or both
- Use the **General Addition Rule**:

$$\boldsymbol{P(A\ or\ B) = P(A) + P(B) - P(A\ and\ B)}$$

Probabilities: a visual summary



Conditional probabilities



The results are in…

		VOTE			
		Rubio Lagos	Marrón	Yamaguchi	
BALLOT	real	1620	1044	936	**3600**
	fake	0	666	234	**900**
		1620	**1710**	**1170**	**4500**

- But unbeknownst to the Comisión Electoral, a lot of fake ballots are cast (using ID cards of members that are already deceased or in prison)
- If we look at all ballots, the winner of the election is Marrón (winner takes all):

$$P(Marrón) = \frac{1710}{4500} = 0.38$$

- But if we only look at the real ones?

$$P(Marrón|real) = \frac{1044}{3600} = 0.29$$

Rubio Lagos has more (45%)!

- This is a case of **statistical dependence**:

$$\boldsymbol{P(B) \neq P(B|A)}$$

  - **When *A* happens, the probability of *B* changes**

How do we explain this dependence?

		VOTE			
		Rubio Lagos	Marrón	Yamaguchi	
BALLOT	real	1620	1044	936	**3600**
	fake	0	666	234	**900**
		1620	**1710**	**1170**	**4500**

- Well, Gracious Griselda's team must have submitted a huge amount of fake ballots. We see a large proportion indeed:

$$P(Marrón|fake) = \frac{666}{900} = 0.74$$

- Makes sense: if support for Marrón depends on whether the ballot is real, it also depends on whether the ballot is fake (the only other option)
- The overall $P(Marrón) = 0.38$ is kind of an average of the conditional probabilities (0.29 and 0.74)

Hideki Yamaguchi tried to cheat his way to victory as well…

		VOTE			
		Rubio Lagos	Marrón	Yamaguchi	
BALLOT	real	1620	1044	936	3600
	fake	0	666	234	900
		1620	**1710**	**1170**	**4500**

- Has it helped him at all?

$$P(Yamaguchi) = \frac{1170}{4500} = 0.26$$

$$P(Yamaguchi|real) = \frac{936}{3600} = 0.26$$

$$P(Yamaguchi|fake) = \frac{234}{900} = 0.26$$

- Looks like he cheated just enough to keep his result the same
- So the events 'real ballot' and 'Yamaguchi vote' are statistically **in**dependent:

$$\boldsymbol{P(B) = P(B|A)}$$

  - **When *A* happens, the probability of *B* stays the same**

The results are in… (continued)

		VOTE			
		Rubio Lagos	Marrón	Yamaguchi	
BALLOT	real	1620	1044	936	**3600**
	fake	0	666	234	**900**
		1620	**1710**	**1170**	**4500**

- Another remarkable fact of this election: there were <u>no</u> fake ballots cast in favour of Rubio Lagos
  - None that fit <u>both</u> events
- When two events can't occur together, they are **disjoint**:

$$\boldsymbol{P(A\ and\ B) = 0}$$

- In that case, it also applies that

$$P(B|A) = 0$$

- Disjoint events are therefore extremely <u>dependent</u>: $P(B) \neq P(B|A)$. After all, for example,

$$P(Rubio\ Lagos) = \frac{1620}{4500} = 0.36 \qquad \text{but} \qquad P(Rubio\ Lagos|fake) = \frac{0}{900} = 0$$

Binomial distributions

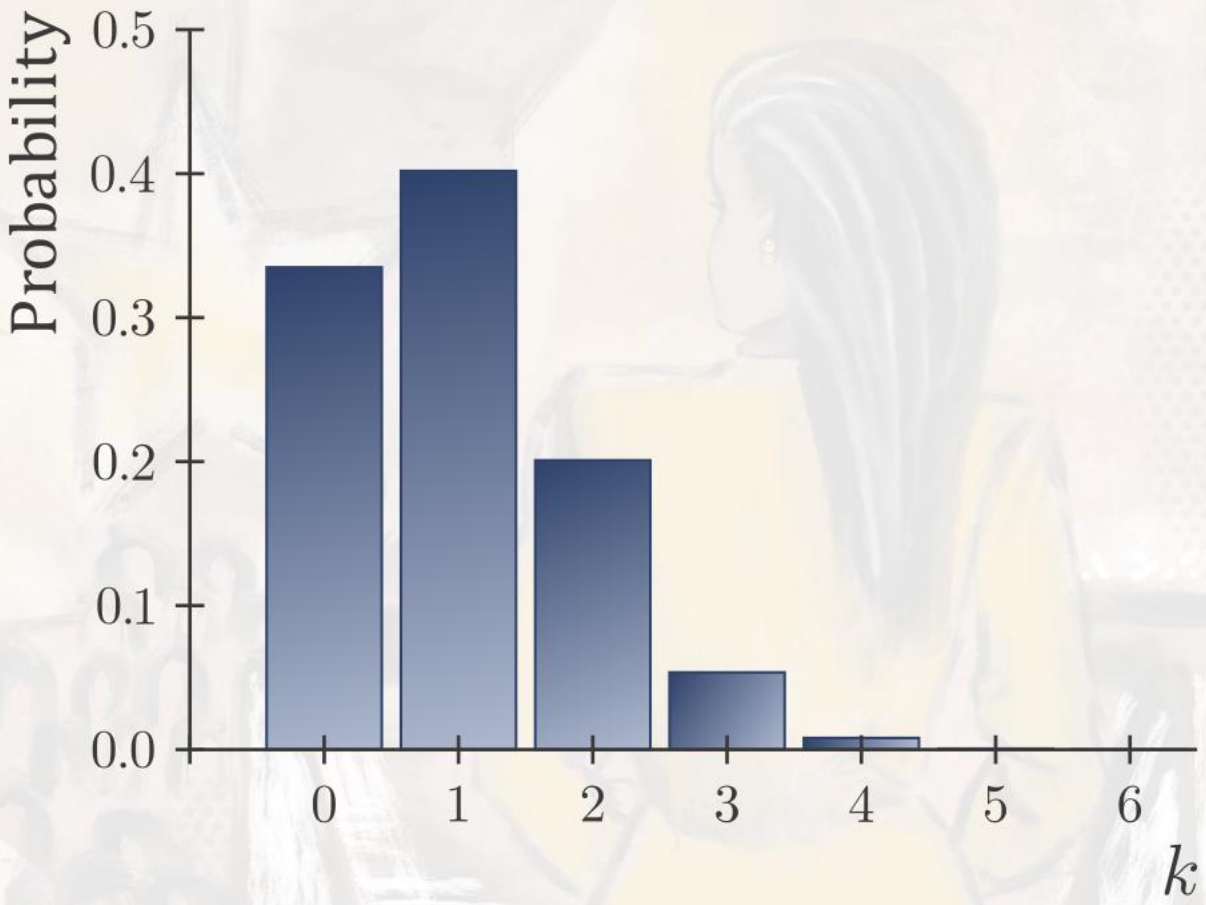
Somebody clean the carpet

At the victory party, Marrón's senior staff plays a game of Russian roulette

- Rules of this variant:
  - Fill a 6-round revolver with a single bullet, then turn the chamber at random
  - Put the gun to your head and pull the trigger
  - If the weapon doesn't go off: give the chamber another whirl and pass the revolver to the next player
  - If the weapon does go off: have someone else load the gun with a new bullet and pass it on the next player
- Thus, for every player, the probability of an untimely end is always 1/6
- n = 6 cartel members decide (after too much vodka) to participate
- What can happen?

The binomial distribution

- The number of players who accidentally shoot themselves is *X*
  - Mathematicians would call this the number of **successes**…
- *X* is a **random variable**
- We call it a **binomial** random variable ('two names')
  - because each individual player will fall into 1 of 2 categories: 'success' (the gun goes off) or 'failure' (it doesn't)
  - and *X* counts how many players belong to the successful category
- In this case, *X* can take on every natural value from 0 to 6
- What's the probability of each of those values?



The binomial distribution: probabilities

- We can extend the <u>multiplication rule</u>!
- For example, the probability of 6 casualties equals

Success *and* success *and* success *and*…

$$P(X=6)=\frac{1}{6}*\frac{1}{6}*\frac{1}{6}*\frac{1}{6}*\frac{1}{6}*\frac{1}{6}=\left(\frac{1}{6}\right)^6$$

- The probability of 5 casualties is really 5 successes *and* 1 failure. Plus, there are several possible *orders* (6 in total)!

$$P(X=5)=\left(\frac{1}{6}\right)^5*\left(\frac{5}{6}\right)^1*6$$

- How can we determine the number of orders quickly? Official notation:

$$\binom{n}{k}$$

- Pronounce as '*n* choose *k*'
- *k* is the value that *X* takes on (the number of successes we observe)

- Example: how many combinations of players yield 5 accidents (and therefore 1 survivor)? As we just said:

$$\binom{6}{5}=6$$

- We call this number of possible combinations the **binomial coefficient**
  - Ask your calculator: use the button *nCr* (type in *6 nCr 5*)

The binomial formula

- There's a pattern to those probability calcs:
  - 0 successes? 6 failures
  - 1 success? 5 failures
  - 2 successes? 4 failures
  - So in general: *k* successes? Then *n* – *k* failures!
- This gets us a general formula for binomial variables:

Probability of *k* successes (*k* casualties)

Total number of players. In general, we call this the *number of trials*

$$\boxed{P(X = k)} = \binom{\boxed{n}}{k} \boxed{p}^{k} \boxed{q}^{n-k}$$

Probability of failure per player (complement rule: $q = 1 - p$)

Probability of success per player (1/6 in this case)

There are many other types of random variables

- It all depends on the question we ask!
- For example: how many attempts will it take for someone to win the game (i.e. make the revolver go off)?
  - Now the number of trials (*n*) is no longer fixed, but the number of successes is (*k* = 1)
  - In other words, the random variable becomes *N*: the **number of trials until a success occurs**
  - We can describe it with the **geometric distribution**:

$$\boldsymbol{P(N = n) = q^{n-1} * p}$$

  - The idea is that you first need $n - 1$ failures (with probability $q = 1 - p$), and then a success (with probability p)
  - Example calculation (*p* = 1/6): the probability that the revolver will go off for the 3rd player (and no earlier) equals

$$P(N = 3) = \left(\frac{5}{6}\right)^2 * \left(\frac{1}{6}\right) = 0.1157$$

Further topics for this week (self-study ☺)

- Sharpe, chapter 5
  - Tree diagrams
- Sharpe, chapter 6
  - Expected values and their properties
  - Standard deviations and their properties
  - Bernoulli trials and discrete probability models

- Keep in mind: we expect you to read the whole book chapter. This slide highlights the most important aspects that were not covered in this lecture

Take a break

After this: mathematics
See you again soon! ☺